Evan Mullins

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EDUCATION

Bachelor of Science in Materials Science and Engineering Expected May 2029
University of Florida, Gainesville, FL **GPA: 4.00/4.00**

- Relevant Coursework: Junior Materials Lab, Mechanics of Materials, Thermodynamics of Materials, Error Analysis & Optimization Methodologies, Inorganic Materials

EXPERIENCE

Carbon Fiber Product Development Intern – Zoltek Corporation May 2026 – August 2026

- Defined conductivity thresholds for CFRTP composites to inform sales team technical claims by designing and executing 400 trials at 8 fiber loading levels across 4 percolation studies per ASTM, ISO, and IEC standards.
- Redesigned the fiber sizing verification process by engineering a multi-flask DCM reflux system with an integrated chilled-water condenser, reducing processing time by 75% and doubling batch capacity.
- Characterized thermal and compositional properties for new product development, historical test verification, and NCR compliance across 20 pilot batches using DMA, TGA, FTIR, and DSC.
- Optimized catalyst type and loading across 20 candidates using gel, viscosity, and exotherm testing across 6 pilot resin systems, establishing pot life benchmarks adopted during pilot cycles.

Composites Development Engineer – Swamp Launch, University of Florida April 2026 – Present

- Constructed and designed CF/fiberglass filament winder for rapidly prototyped composite airframes and COPVs in alignment with Swamp Launch's 5-year Spaceshot Development team.
- Reduced winder time by 50%, validated across 5 test runs, by rebuilding Python software to automate G-code generation.
- Developed ASTM-based lap shear test protocol for structural foam/prepreg bonding across 2 systems to optimize fin core compatibility and composite quality.

IREC Composites Engineer – Swamp Launch, University of Florida August 2025 – Present

- Manufactured 10+ SRAD composite parts per cycle (airframes, fins, couplers, nosecone), controlling epoxy formulation, vacuum cure, and post-processing to reduce defects and decrease layup time by 50%.
- Reduced competition fin thickness by 48% while maintaining 80% structural integrity determined by a student-designed flutter safety calculator after executing the team's first tip-to-tip layup trial and prepreg/autoclave sandwich panel cure.

Undergraduate Research Assistant – MONSTER Group, University of Florida August 2026 – Present

- Began computational thermodynamics research using PyCalphad, an open-source CALPHAD framework, to model phase diagrams ahead of UF's sunset of its commercial phase-diagram software.
- Authored onboarding documentation and training materials for incoming undergraduate researchers adopting PyCalphad.

PROJECTS

HAB Payload Engineer – NASA Florida Space Grant Consortium, KSC Visitor Complex July 2026 – August 2026

- Designed and optimized material selection and payload design to analyze convective heat transfer collapse of aluminum components during 100,000 ft flight, validated via comparing Python model against operating temperature constraints.
- Optimized BOM across 3 payloads within a 4lb, $800 budget, coordinating lead times to greenlight all experiments for flight.
- Selected by FSGC to produce audiovisual documentation for content briefed to Florida's congressional delegation, Space Trek websites, and social media of 2,000 followers.

LEADERSHIP

Outreach Director – Freshman Leadership in Engineering, University of Florida April 2026 – Present

- Oversaw 10 freshmen and organized monthly projects averaging 20 attendees, streamlining logistics, communication, and finances as Outreach Director for a program selecting from 800+ applicants.
- Led "Pack it Pink" fundraiser for breast cancer patients, generating $2,613 in 12 days (program record).

SKILLS

Manufacturing & Lab: DOE, ANOVA, Regression Modeling, Autoclave Curing, RTM, Vacuum Infusion, NDT (Ultrasonic), Instron Testing, TGA, DMA, DSC, FTIR, 5S Lean Principles, 3D Printing, Failure Analysis, Soldering
Engineering Software: Fusion360, CES Selector, OpenRocket, JMP, ImageJ, OpenSim
Languages: Python (sklearn, scipy, numpy, matplotlib, pandas), G-code, MATLAB, HTML, JavaScript
Interests: Jazz Saxophone, Filmmaking, Free Verse Poetry, Volleyball, Foreign Languages